REVISION HISTORY

Schematics Index:

Revision	Description	Date	Drawn	Checked
Ver 1.0	Initial	2015-02-06		

ORANGE_PI_PC_Ver 1.0 Initial 2015-05-29

ORANGE_PI_PC修改内容：1、删除WIFI模块及相应电源模块
2、删除U12（USB HUB）、USB2模块及相应电源模块
3、删除摄像头部分的电源模块
4、删除按键SW2、SW3按键
5、删除部分NC电容电阻
6、添加侧插USB插座P1以及相应电源模块

ORANGE_PI_PC_Ver 1.1 Initial 2015-06-23

ORANGE_PI_PC_V1.1修改内容：1、更改了复位电路
2、更改了电源模块CPUX，RTC电源与AVCC电源模块合并为一路
3、删除SGM89000模块（CVBS部分）
4、降压前5V：HDMI、CSI、USB、VDD-CPUX、EXT Port；
降压后5V：VCC-DRAM、1.2V、3.3V、AVCC

ORANGE_PI_PC_Ver 1.2 Initial 2015-07-10

ORANGE_PI_PC_V1.2修改内容：1、AVCC电源模块使能脚直接接5V
2、替换了电源模块CPUX（删除：U53(SY8089A)，22uf电容C307、C186；10uf电容C184、C308；
0.1uf电容C183、C185、C181；2.2uh电感L4；
10K电阻R233、R239；11.8K电阻R235；30.1K电阻R237；0R电阻R238；三极管BSN20，Q5）
（增加：U9（SY8106A），22uf电容C67、C72、C73；10uf电容C310、C68、C74；104电容C66、C71；100nf电容C64；2.2uh电感L4；
44.2K电阻R31，10nf电容C78；2.2uf电容C77，100K电阻R30、R27，220pf电容C75）
3、HDMI、USB的ESD器件全部NC
4、DRAM电源改为1.35V，R14使用120K电阻

ORANGE_PI_One_Ver 1.0 Initial 2015-12-8

ORANGE_PI_One_V1.0修改内容：1、删除32晶振
2、整体更换1.2VC电源模块
3、删除双层USB模块/删除单USB的电源模块，改用电源5V直接供电，增加滤波电容C86、C309/单USB信号选用DP1、DM1
4、删除耳机和MIC
5、HDMI增加ESD器件
6、去掉红外

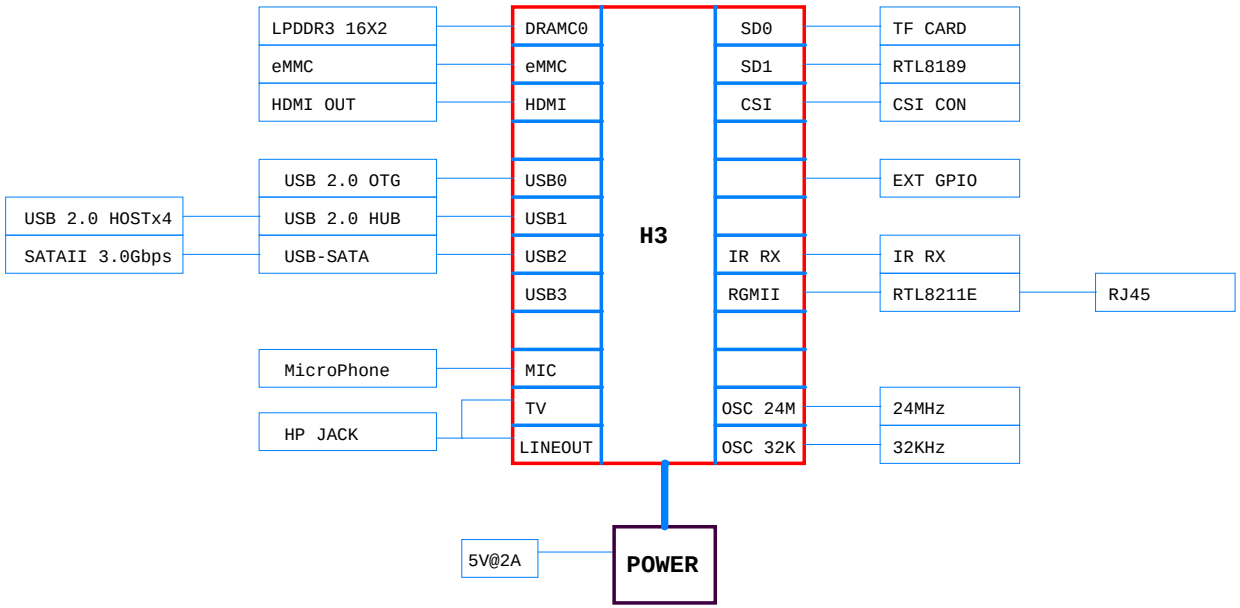
ORANGE_PI_Lite_Ver 1.0 Initial 2015-12-24

ORANGE_PI_Lite_V1.0修改内容：1、增加R20
2、增加P2，USB模块
3、删除EMAC，增加WIFI，增加WIFI电源模块

ORANGE_PI_Lite_Ver 1.1 Initial 2016-04-09

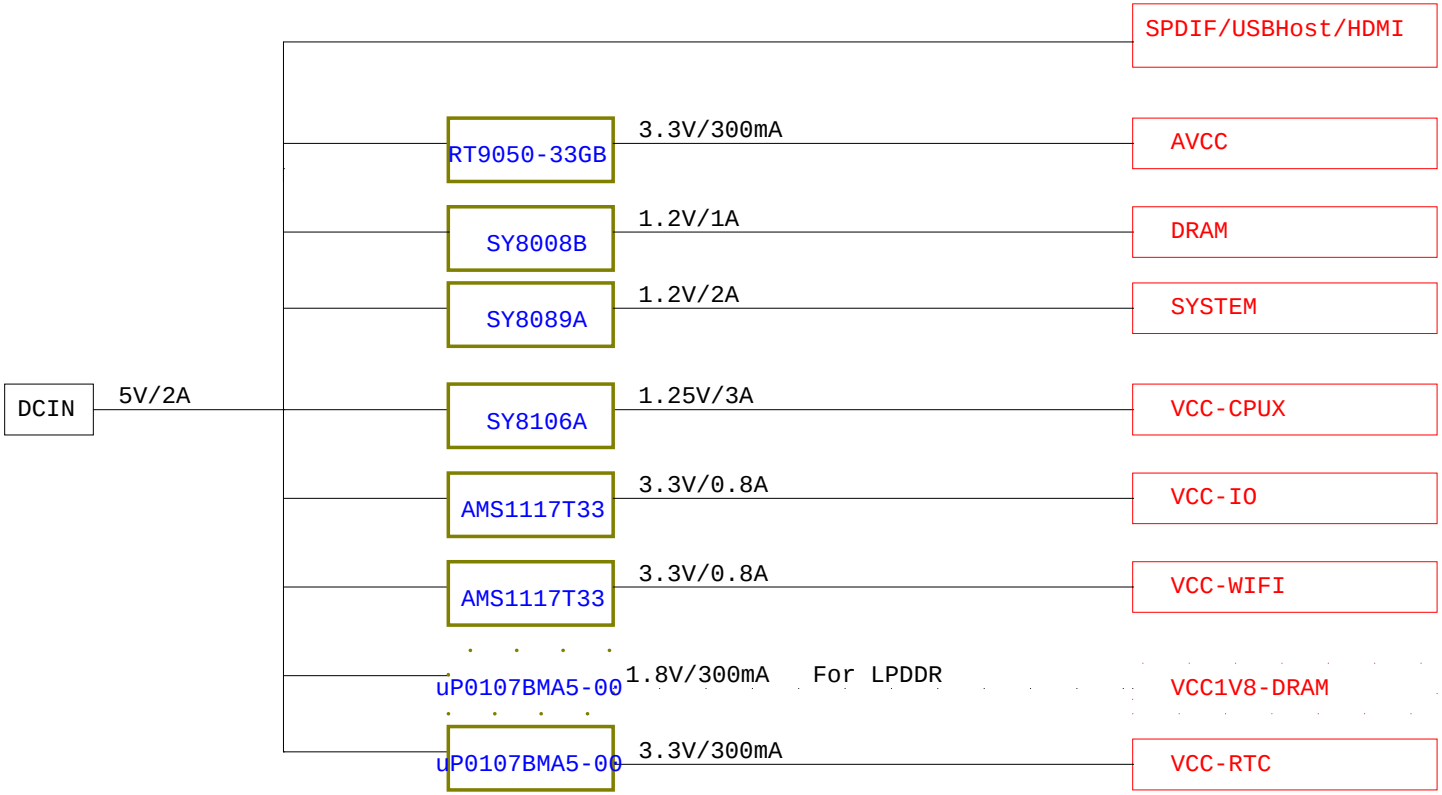
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BLOCK



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POWER TREE



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GPIO ASSIGNMENT

PIN	Define	CFG	Function
PA0	TMS/DRVVBUS0	3/1	JTAG /USB
PA1	TCK/DRVVBUS1	3/1	
PA2	TDO/WPS	3/1	
PA3	TDI	3	
PA4	UART-TX	3	UART
PA5	UART-RX	3	
PA6	NC	7	
PA7	NC	7	
PA8	NC	7	
PA9	NC	7	
PA10	NC	7	
PA11	NC	7	
PA12	NC	7	
PA13	NC	7	
PA14	NC	7	
PA15	STATUS-LED	1	LED
PA16	MUTE	1	AV
PA17	SPDIF-OUT	2	SPDIF
PA18	NC	7	
PA19	NC	7	
PA20	NC	7	
PA21	NC	7	

PIN	Define	CFG	Function
PC0	NWE	2/3	NAND /eMMC /NOR
PC1	NALE	2/3	
PC2	NCLE	2/3	
PC3	NCE1	2/3	
PC4	NCE0	2	
PC5	NRE	2/3	
PC6	NRB0	2/3	
PC7	NRB1	2	
PC8	NDQ0	2/3	
PC9	NDQ1	2/3	
PC10	NDQ2	2/3	
PC11	NDQ3	2/3	
PC12	NDQ4	2/3	
PC13	NDQ5	2/3	
PC14	NDQ6	2/3	
PC15	NDQ7	2/3	
PC16	NDQS	2/3	

PIN	Define	CFG	Function
PD0	NC	7	
PD1	NC	7	
PD2	NC	7	
PD3	NC	7	
PD4	NC	7	
PD5	NC	7	
PD6	NC	7	
PD7	NC	7	
PD8	NC	7	
PD9	NC	7	
PD10	NC	7	
PD11	NC	7	
PD12	NC	7	
PD13	NC	7	
PD14	NC	7	
PD15	NC	7	
PD16	NC	7	
PD17	NC	7	

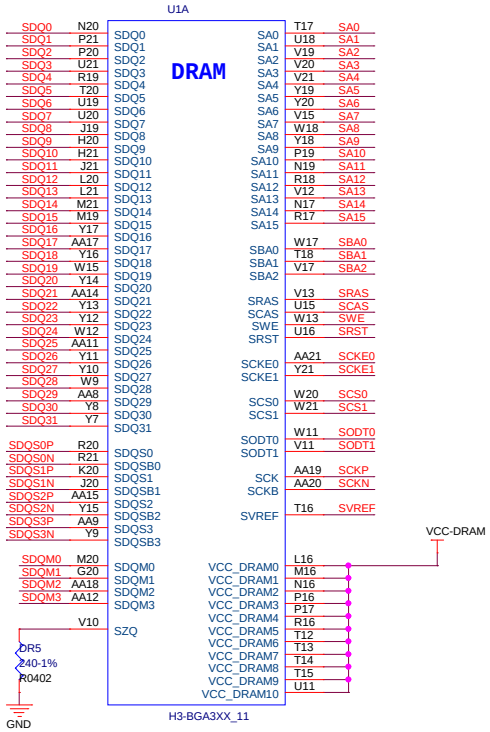
PIN	Define	CFG	Function
PE0	NC	7	
PE1	NC	7	
PE2	NC	7	
PE3	NC	7	
PE4	NC	7	
PE5	NC	7	
PE6	NC	7	
PE7	NC	7	
PE8	NC	7	
PE9	NC	7	
PE10	NC	7	
PE11	NC	7	
PE12	NC	7	
PE13	NC	7	
PE14	NC	7	
PE15	NC	7	

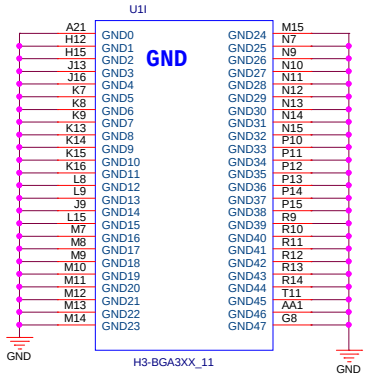
PIN	Define	CFG	Function
PF0	D1	2	CARD0
PF1	D0	2	
PF2	CLK	2	
PF3	CMD	2	
PF4	D3	2	
PF5	D2	2	
PF6	DET	0	

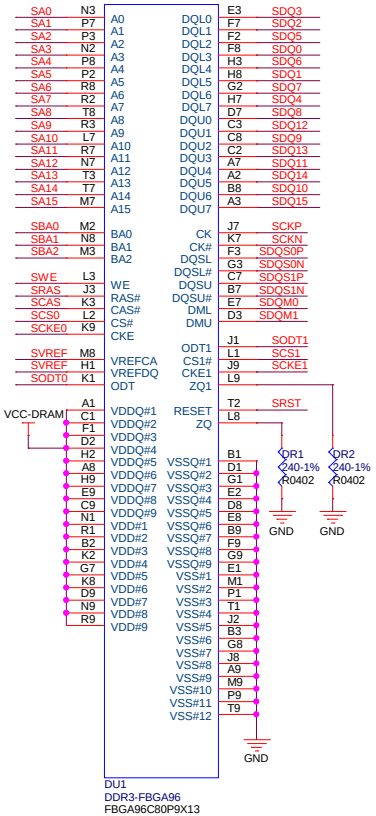
PIN	Define	CFG	Function
PG0	NC	7	
PG1	NC	7	
PG2	NC	7	
PG3	NC	7	
PG4	NC	7	
PG5	NC	7	
PG6	NC	7	
PG7	NC	7	
PG8	NC	7	
PG9	NC	7	
PG10	NC	7	
PG11	NC	7	
PG12	NC	7	
PG13	NC	7	

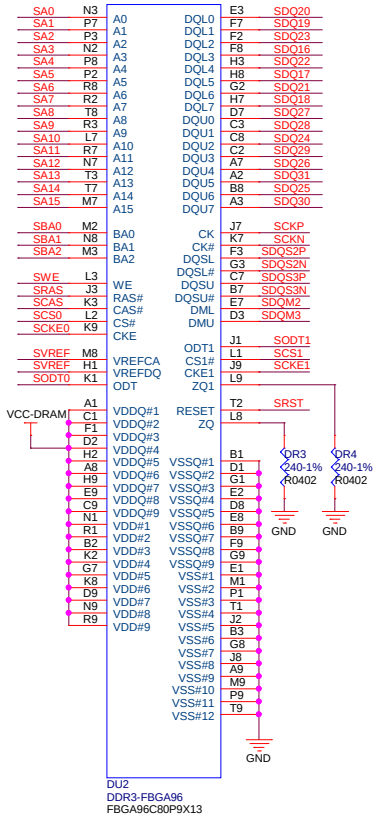
PIN	Define	CFG	Function
PL0	TWI	2	TWI
PL1	TWI	2	
PL2	USB0-DRVVBUS	1	USB
PL3	USB1-DRVVBUS	1	
PL4	RECOVERY	0	KEY
PL5	VCC-IO-EN	1	IO-EN
PL6	NC	7	
PL7	WIFI-EN	7	WIFI-EN
PL8	PWR-STB	1	
PL9	PWR-DRAM	1	
PL10	PWR-LED	1	
PL11	IR-RX	2	

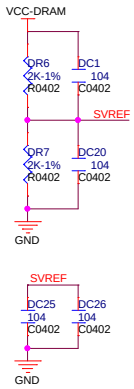
DDR3 16x2

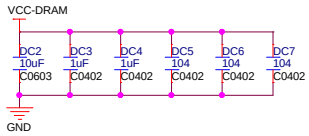


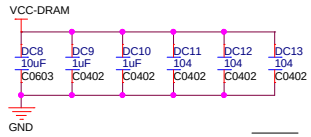


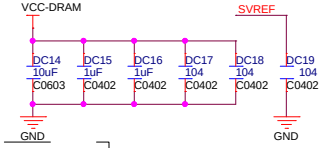


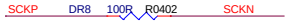


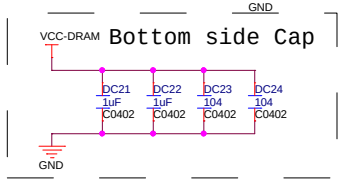


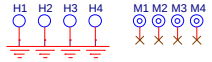






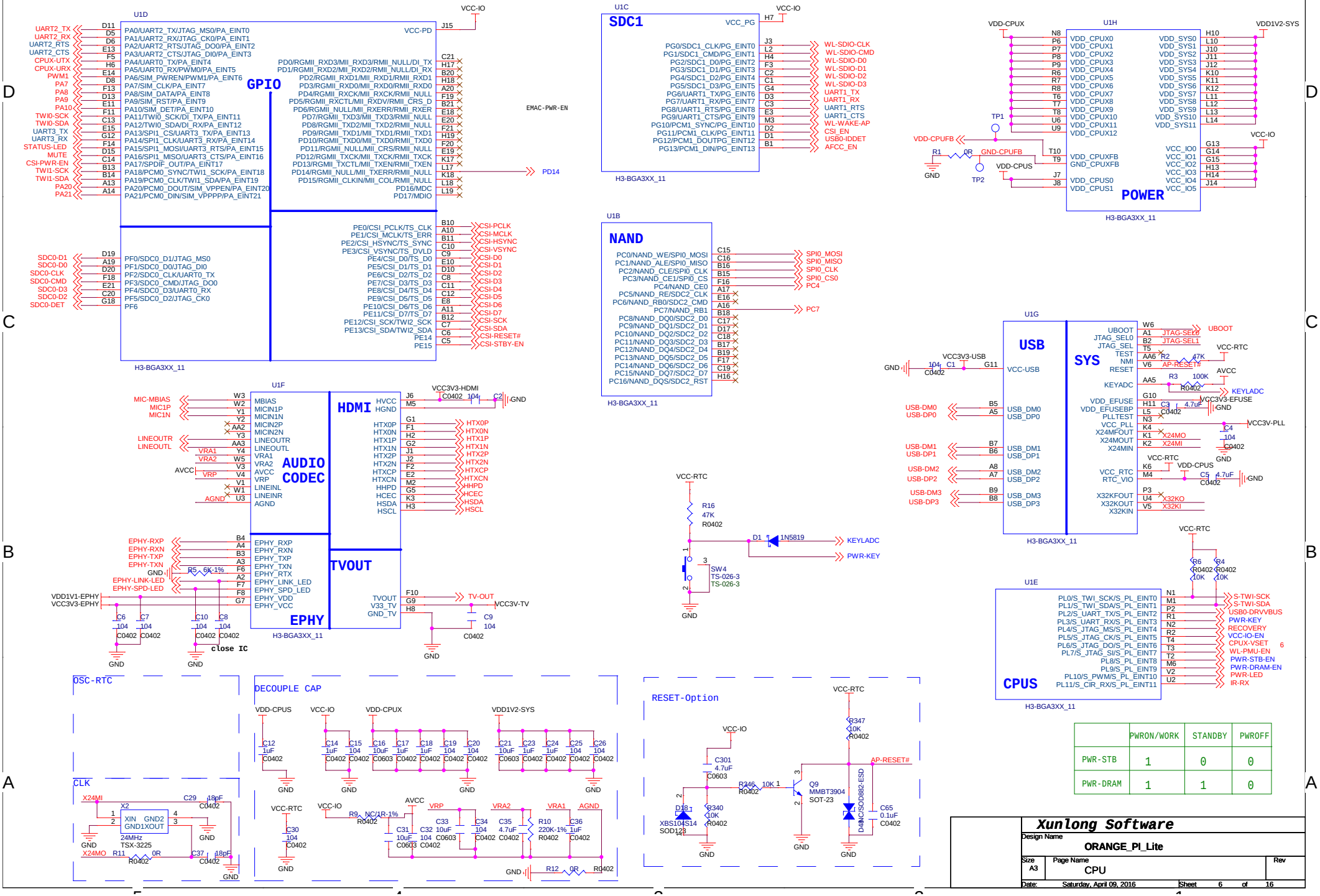






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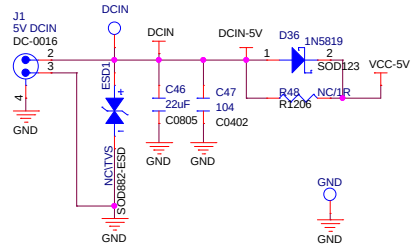
CPU



	PWRON/WORK	STANDBY	PWROFF
PWR-STB	1	0	0
PWR-DRAM	1	1	0

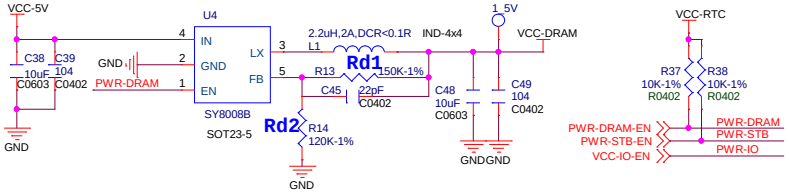
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DCIN

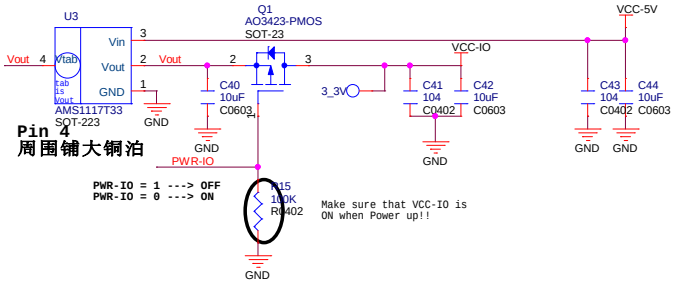


POWER

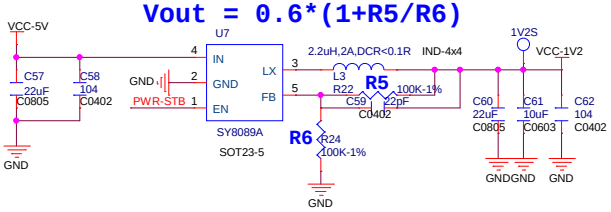
Vout = 0.6*(1+Rd1/Rd2)
VDRAM=1.5V/1A,R2=100K-1%
VDRAM=1.35V/1A,R2=120K-1%



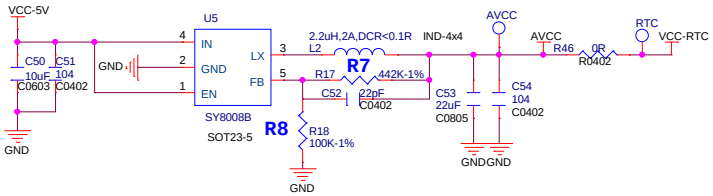
VCCIO 3.3V/1A



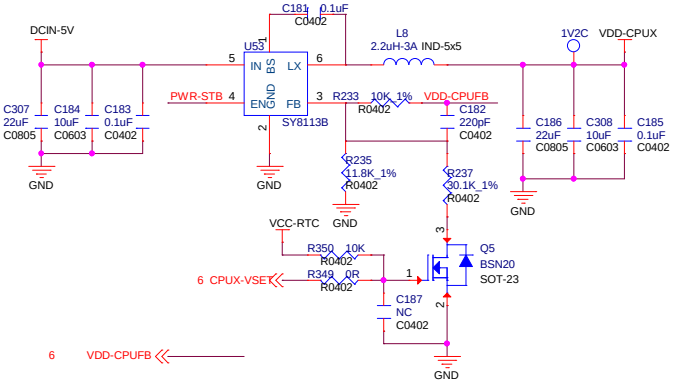
SYSTEM 1.2V/2A
Vout = 0.6*(1+R5/R6)



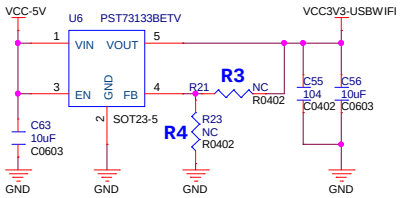
AVCC 3.3V/1A



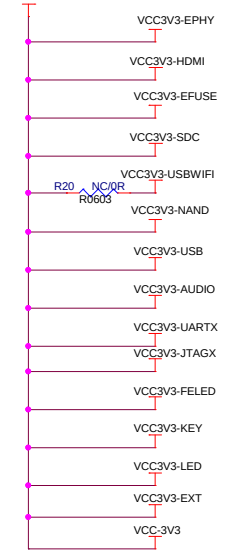
CPUX 1.2V/6A
Vout = 0.6*(1+R1/R2)



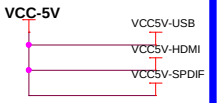
WIFI Power 3.3V/300mA



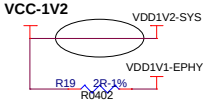
VCC-IO



VCC-5V



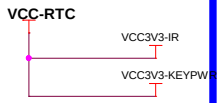
VCC-1V2



VDD-CPUX

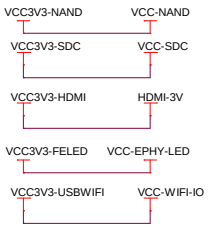
VCC-DRAM

VCC-RTC



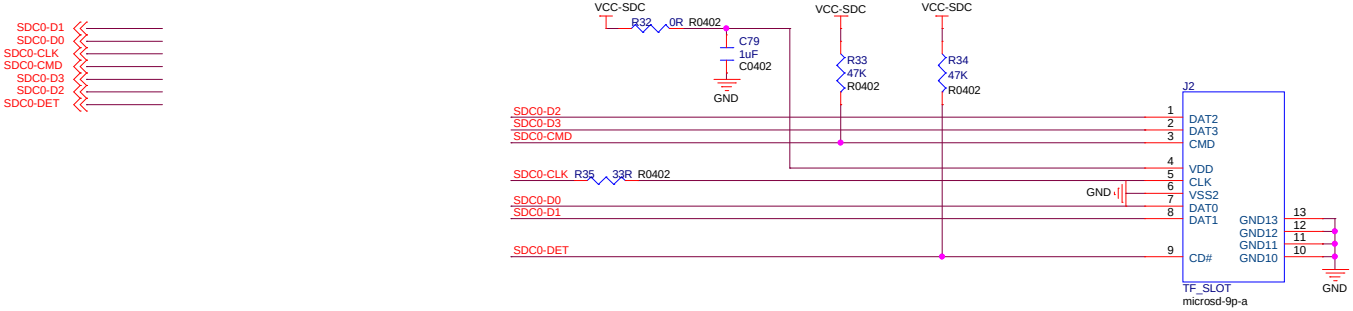
AVCC





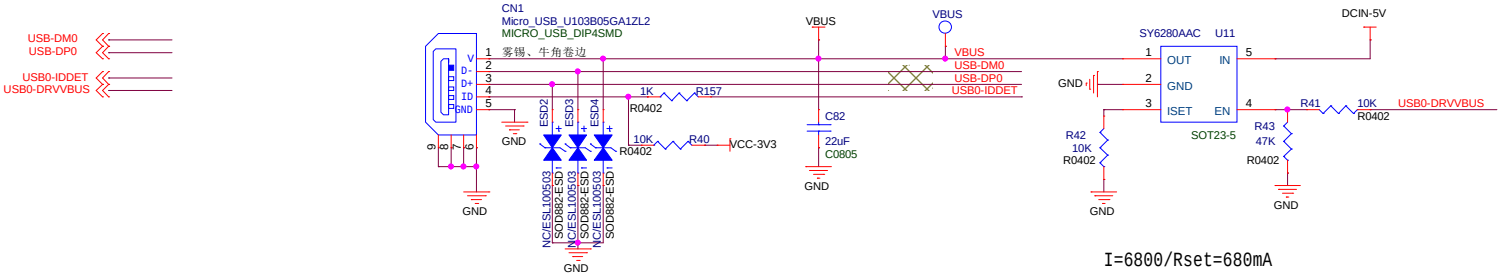
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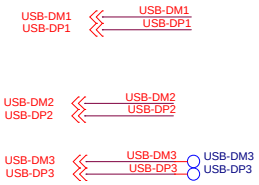
TF-eMMC



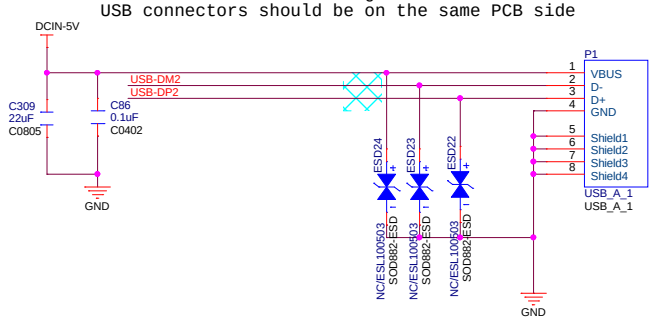
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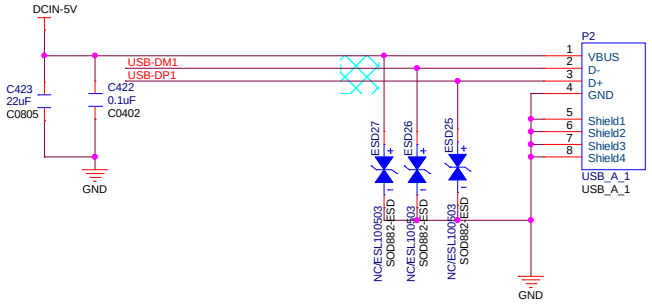
USB





note: Make sure the routing between the ESD and the USB connectors should be on the same PCB side





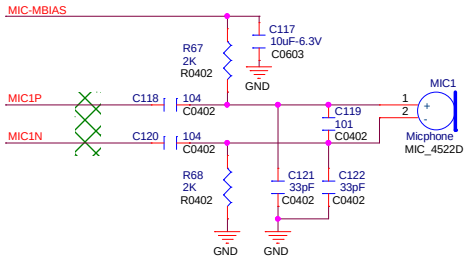
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AV-MIC-IR

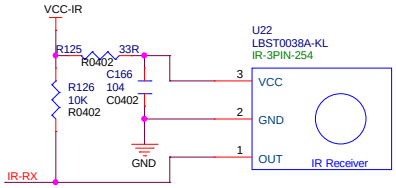






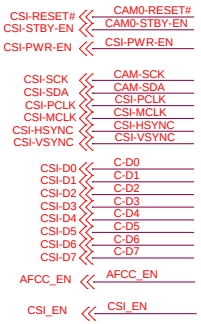


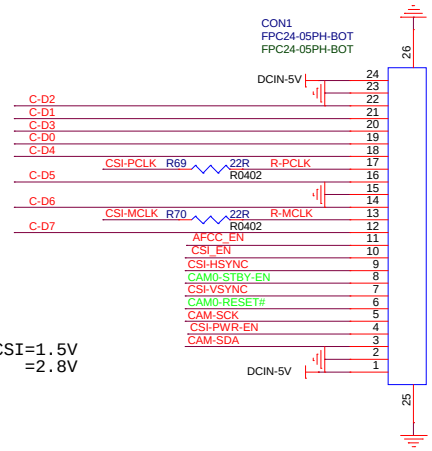
IR



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Camera

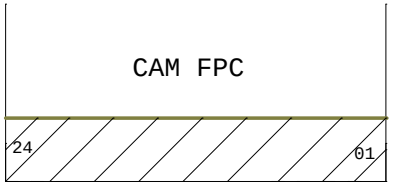


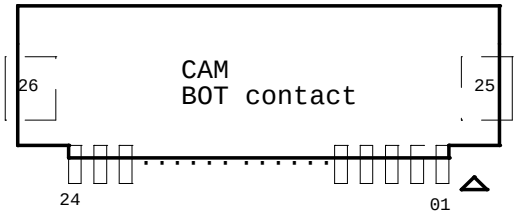


VDD1V5-CSI=1.5V
VCC-CSI =2.8V



Close to AP



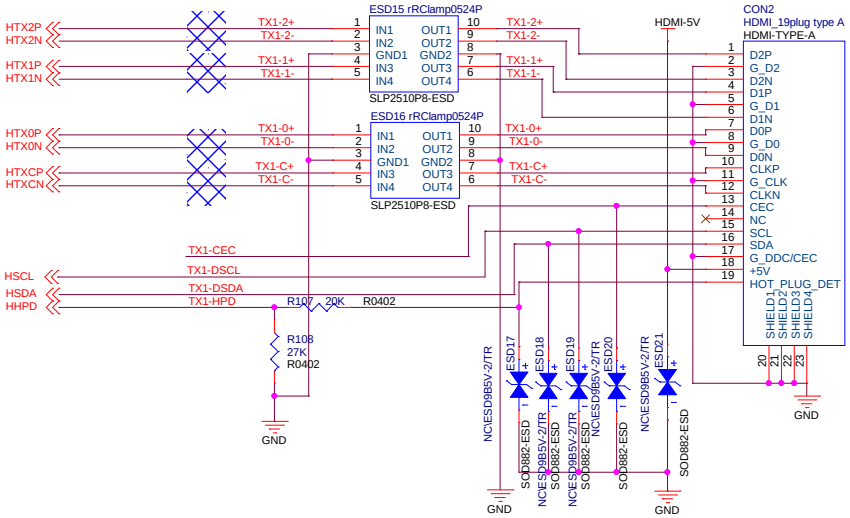


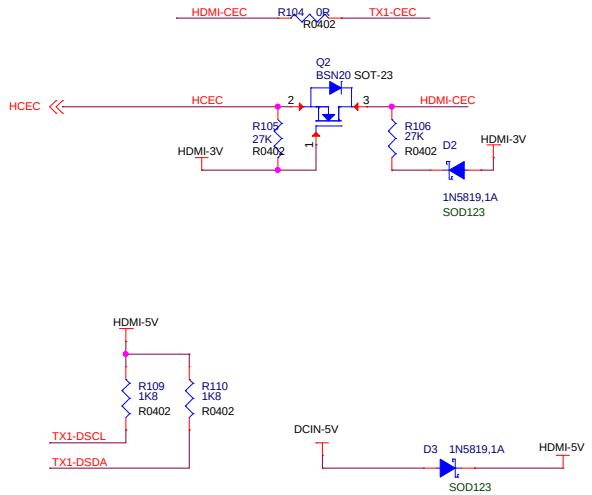
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EMAC

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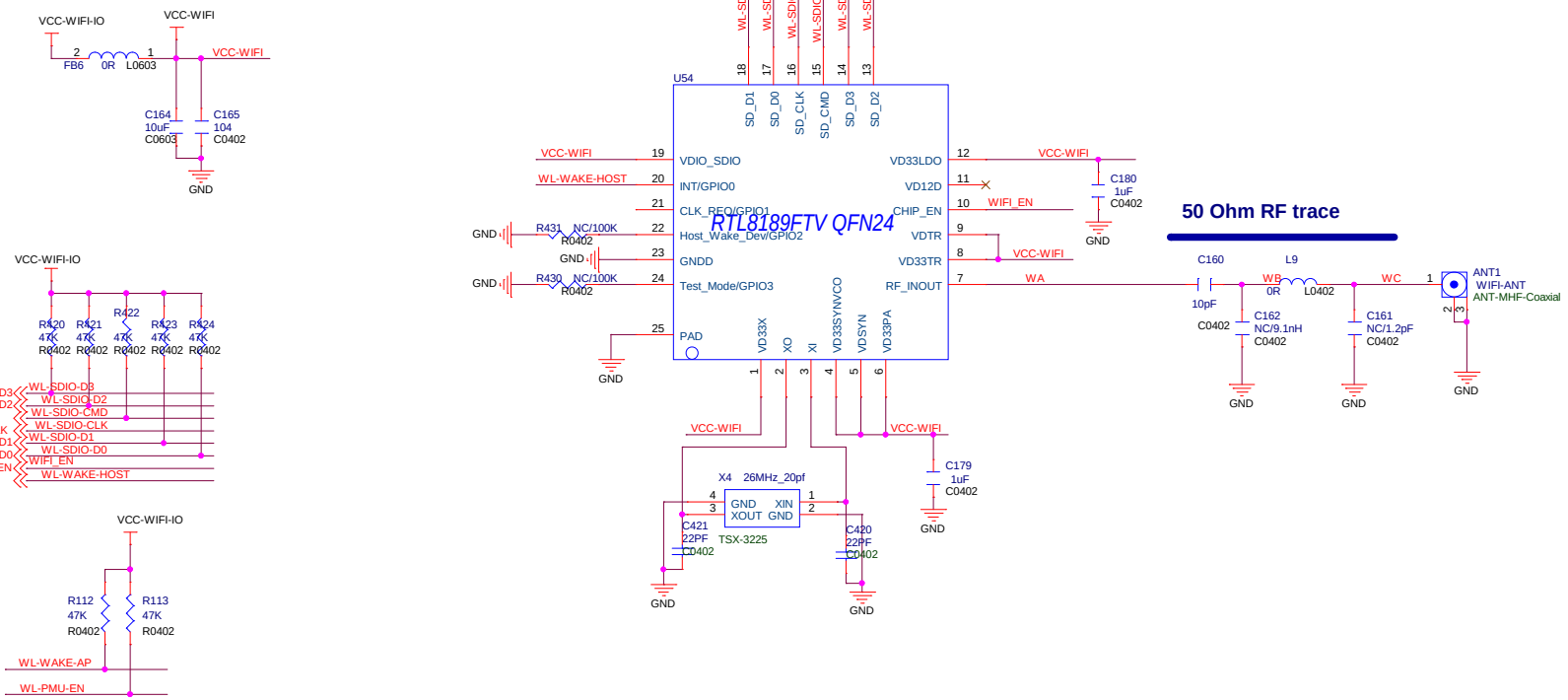
HDMI





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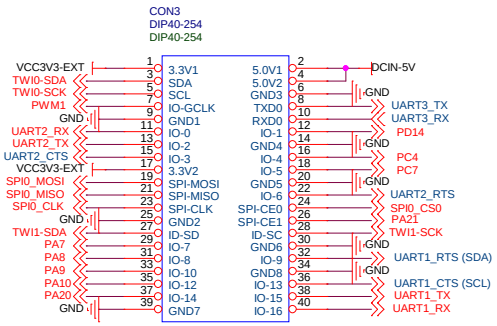
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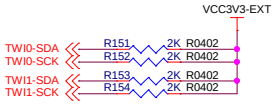


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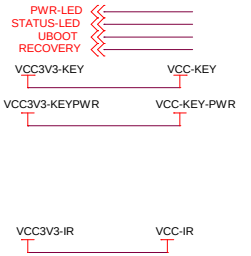
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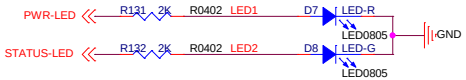
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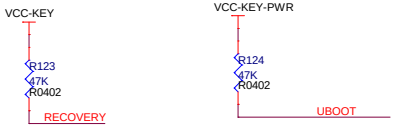


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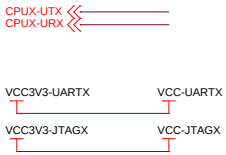


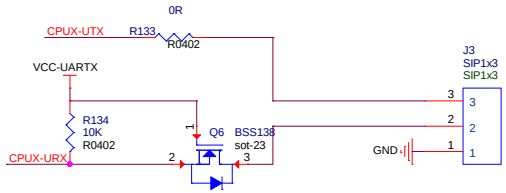
KEY



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DEBUG





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